

Decision Analysis and Impact Pathway for Agroforestry Integration on Dairy Farms in the Eifel Region, Germany

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Agroforestry for Dairy Farms

We showcase the application of holistic decision supporting approaches in supporting agroforestry interventions for dairy farms in the German Eifel.

Agroforestry on Dairy farms

- Above 25 °C and 50% humidity dairy cows can suffer from heat stress, negatively affecting animal health and milk yield (Gauly et al., 2013).
- Agroforestry has the potential to reduce heat stress and radiation effects on dairy cows (Kendall et al. 2006).

Agroforestry on dairy farms in the Eifel Region

- In the Eifel, hedges of beech trees, maple trees, and other deciduous trees are traditionally used as windbreakers, natural fences, and for fire wood.
- Farmers in the Eifel Region are open to ecologically sustainable methods of farming (LWK NRW 2013). Agroforestry may be a scalable option for dairy farmers in the region.

Supporting farm-level decisions

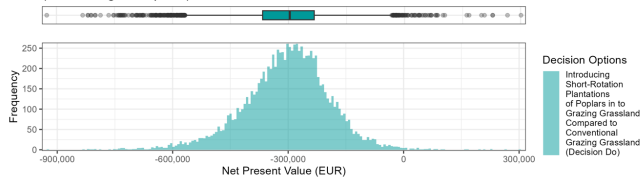
Decision sciences offer methods for incorporating expert knowledge to support complex decisions in agricultural development (Luedeling & Shepherd 2016).

Should farmers be willing to establish agroforestry?

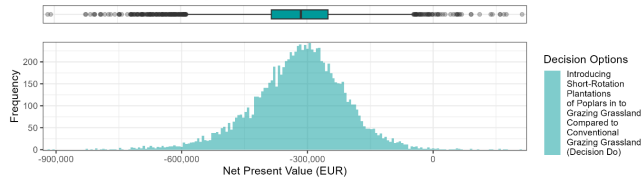
- We defined interventions and uncertainties of agroforestry interventions together with dairy farmers in the Eifel.
 - We simulate a 30-year intervention of grazing cows on grassland under short-rotation poplar trees compared to grassland without trees (10 hectares with north-south oriented rows).
- We ran a Monte Carlo Simulation (MC) with 10,000 runs to simulate the decision options and calculate the Net Present Value (NPV).
- Based on the MC results, we calculate the Variable Importance in the Projection (VIP) from a Projection to Latent Structure (PLS) to assess the sensitivity of the NPV output with the many collinear model inputs.
- We calculated annual cashflows for dairy farmers over the 30 years

Model outputs

Distribution of Net Present Values of Introducing Agroforestry System (with Ecological Aspects)



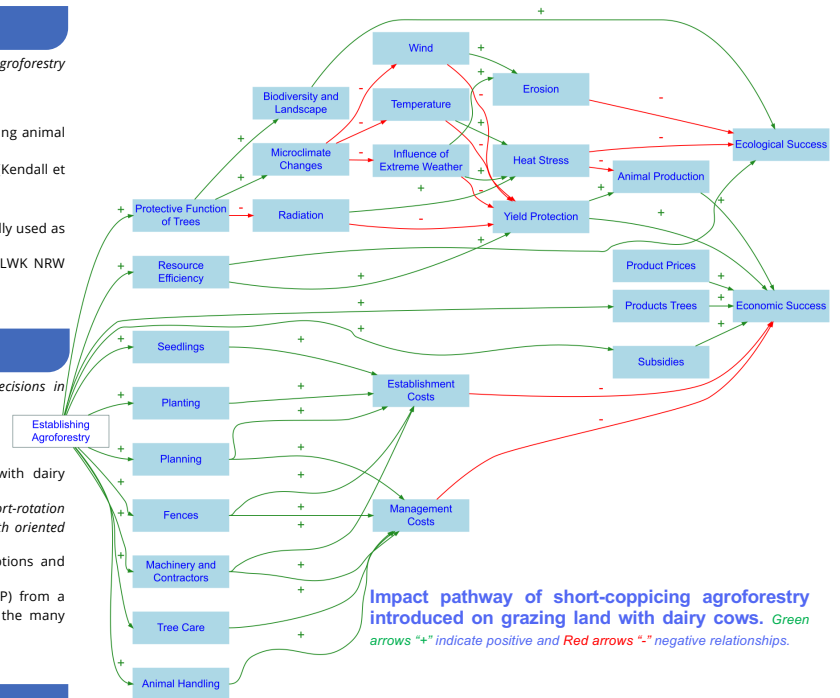
(without Ecological Aspects)



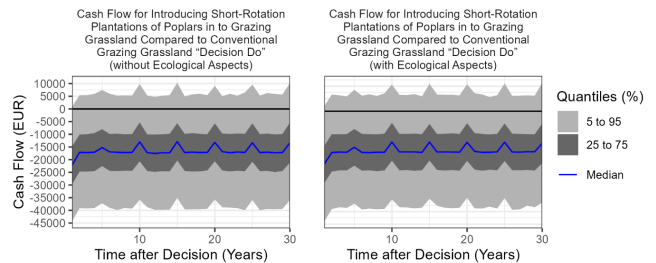
NPV of agroforestry with and without consideration of the ecological implications showed general losses for the farmer. The histograms represent the density of the 10,000 results of the MC simulation model.

Implications for Agroforestry

- Defining models with farmers ensures that our research is grounded in their practical experiences and needs.
- Given our model assumptions, this AF intervention is not economically feasible in the Eifel.
- VIP indicates that important factors that impact the decision:
 - Farmers have a time sensitivity to returns (discount)
 - The proportion of tree coverage can influence grass yield and ultimately milk yield
- Future research should consider other agroforestry interventions for dairy farms, given the potential for improving animal health.

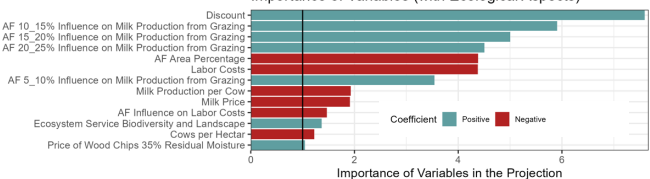


Impact pathway of short-coppicing agroforestry introduced on grazing land with dairy cows. Green arrows "+" indicate positive and Red arrows "-" negative relationships.



Cash Flow for the decision to introduce the agroforestry system of short rotation poplar rows and grazing cows compared to keeping the conventional grazing system for 10.000 model runs

Importance of Variables (with Ecological Aspects)



Variable Importance in the projection for the decision to introduce the agroforestry system of short rotation poplar rows and grazing cows or keeping the conventional grazing system for 10.000 model runs

References and Acknowledgments

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